Total No. of Questions : 4]

SEAT No. :

PF228

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APR-26/SE/Insem-282

S.E. (AI & ML) (Insem)

FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

(2019 Pattern) (Semester - IV) (218553)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates:

1) *Answer Q.1 or Q.2 and Q.3 or Q.4.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Figures to the right indicate full marks.*
4) *Assume suitable data if necessary.*

Q1) a) What do mean by Intelligent Agent? Explain the structure of agent in environment with suitable diagram? **[6]**

b) What is PEAS? **[4]**

c) Differentiate between Artificial Intelligent and Machine Learning. **[5]**

OR

Q2) a) What is AI problem? Describe any two AI problems. **[5]**

b) What is meant by AI Turing test? Explain with suitable example **[5]**

c) What is the evolution of artificial intelligence (AI) over time, and what are some of the major applications of AI in various fields and industries? **[5]**

Q3) a) Use the best-first search algorithm to navigate a maze and justify why it is suitable for this type of problem compared to uninformed search methods. **[8]**

b) Implement an online search strategy for a robot navigating an unknown environment and justify the choice of algorithm used. **[7]**

OR

Q4) a) Illustrate how problem reduction can be used to simplify a complex problem like task scheduling in a multi-core processor system. **[7]**

b) Apply a search method with nondeterministic actions to a self-driving car scenario, explaining how it deals with partial observations. **[8]**

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